
Global Food & Nutrition Dashboard

Project Proposal

COMP5120 — Data Visualization · Spring 2026

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GitHub: github.com/miinguyen/global-food-nutrition-dashboard

Motivation & Core Questions

Why It Matters

- ▶ 800 M+ go hungry while 1 B+ are obese — often in the *same* countries.
- ▶ Data exists across OWID, WHO, and Open Food Facts, but is scattered and hard to compare.
- ▶ No unified interactive tool lets users explore diet, health, and product quality together.

Three Core Questions

1. **“What does the world eat?”**
Calorie trends across 180+ countries over six decades.
2. **“How healthy is it?”**
Linking diet to obesity, malnutrition, and life expectancy.
3. **“What’s really in our food?”**
8 000+ products by Nutri-Score & NOVA.

Our solution: A three-tab **Python Shiny** dashboard with 12 interactive charts and 3 ML models — one tab per question.

Initial Project Plan

Deliverables

- ▶ Automated data pipeline (7 raw sources → 3 clean tables)
- ▶ Interactive Shiny dashboard with 3 tabs, 12 Plotly charts
- ▶ ML: K-means, obesity regression, Nutri-Score classifier
- ▶ Final report, data dictionary, and presentation

Timeline

May 18	Proposal submission
May 25	Data pipeline complete
Jun 1	Dashboard MVP
Jun 7	Final submission

Nguyen Thi Tra My

- ▶ Tab 1 & 2 visualizations (Charts 1–8)
- ▶ K-means clustering & PCA
- ▶ Obesity regression model
- ▶ Data pipeline & merging

Tran Thi Hoai Phuong

- ▶ Tab 3 visualizations (Charts 9–12)
- ▶ Nutri-Score ML classifier
- ▶ Dashboard UI & styling
- ▶ Documentation & report

Dataset Description

Q1: What does the world eat?

Sources: OWID Calorie Supply & Food Composition

Scope: 192 countries, 1961–2023

Columns: kcal/capita/day, 26 food groups, 8 macro categories, life exp.

Output: country_yearly
(10 417 rows $\times$ 39 cols)

Q2: How healthy is it?

Sources: WHO GHO — Adult Obesity & Child Underweight

Scope: 200 countries, 1990–2023

Columns: Obesity %, underweight %, 95% CIs, WHO region

Output: country_health
(7 777 rows $\times$ 10 cols)

Q3: What's in our food?

Source: Open Food Facts API

Scope: 8 057 products

Columns: Nutri-Score (A–E), NOVA (1–4), energy, fat, sugar, salt, protein, fiber

Output: products
(8 057 rows $\times$ 15 cols)

Integration: All country sources linked via **ISO-3166 codes** and joined on (iso3, year). Obesity sourced exclusively from WHO (with CIs) to avoid redundancy. All data exported as Parquet for fast loading.

Data Pipeline & Table Relationships

Processing Pipeline

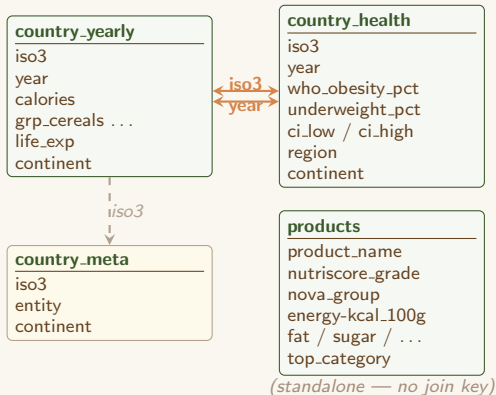
6 Raw Sources:

- ▶ OWID Calorie Supply (13K rows)
- ▶ OWID Food Composition (13K rows)
- ▶ OWID Life Expectancy (21K rows)
- ▶ WHO Adult Obesity (6.5K rows)
- ▶ WHO Child Underweight (2.2K rows)
- ▶ Open Food Facts (10K products)

⇓ data/processing.py ⇓

country_yearly	10,417	rows × 39 cols
country_health	7,777	rows × 10 cols
products	8,057	rows × 15 cols
country_meta	192	countries

Exported as **Parquet** + CSV



Planned Visualizations

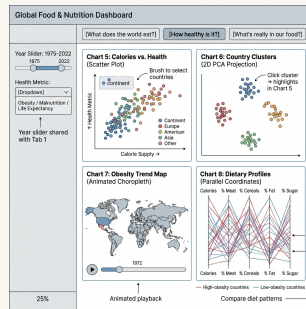
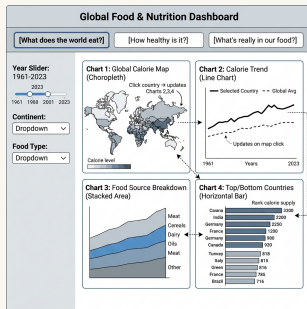
12 Charts (Plotly)

Tab 1 — Diet Patterns

- ▶ Choropleth map (calorie supply)
- ▶ Line chart (country trends)
- ▶ Stacked area (food composition)
- ▶ Horizontal bar (country ranking)

Tab 2 — Health Outcomes

- ▶ Scatter (calories vs. health)
- ▶ 2D cluster projection (PCA)
- ▶ Animated choropleth (obesity)
- ▶ Parallel coordinates (profiles)



Interactivity: Cross-filtering, year sliders, animated playback, click-to-drill, brushing, hover tooltips.

Wireframes: Tab 1 "What does the world eat?" & Tab 2 "How healthy is it?"

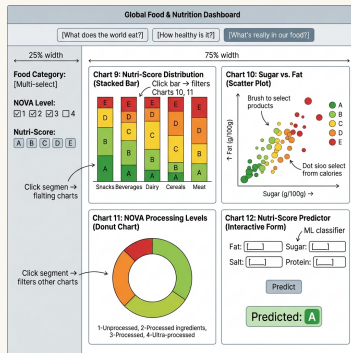
Analytical & ML Methods

ML Models

1. **K-means + PCA**
Cluster countries by dietary profile; visualize in 2D projection.
2. **Obesity Regression**
Predict obesity from food-group calorie shares.
3. **Nutri-Score Classifier**
Predict grade A–E from macros; interactive form in Tab 3.

Tech stack:

Python Shiny · Plotly · scikit-learn · Pandas · Parquet



Tab 3: Nutri-Score distribution (stacked bar) · Sugar vs. Fat scatter (by grade) · NOVA levels (donut) · ML predictor form (macros → grade)

Summary & Questions

What We Will Deliver

- ▶ A **three-tab interactive dashboard** answering what the world eats, how healthy it is, and what's in our food.
- ▶ **12 Plotly charts** with cross-filtering, brushing, and animated playback.
- ▶ **3 ML models** embedded in the dashboard (clustering, regression, classification).
- ▶ Reproducible pipeline: 7 raw sources → 3 Parquet tables.

Roadmap

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GitHub Repository

[github.com/miinguyen/
global-food-nutrition-dashboard](https://github.com/miinguyen/global-food-nutrition-dashboard)

Thank you!

Questions?